



Cookies Are Not the Whole Story: Discrepancies Between Tracking Disclosures and Tracking Behavior

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Cookie banners have become the primary mechanism by which websites obtain consent for tracking and the processing of personal data. Consequently, the disclosures presented through these banners form an important basis for establishing informed consent. Under regulations such as GDPR, consent is valid only when it is informed and specific, requiring websites to clearly disclose essential information, including the type of data being processed. While prior work has mainly focused on banner design choices and stated processing purposes, less is known about whether cookie banners provide sufficient disclosure regarding the actual tracking methods deployed by websites. Given the role of such disclosures in supporting informed consent, we conduct a two-part study to examine the discrepancy between what banners disclose and the tracking behaviors observed on websites. First, using Large Language Models (LLMs), we analyze 33,946 cookie banners from the Tranco Top 100K domains to assess whether tracking is described clearly or vaguely, with 96.8% accuracy. Second, we measure tracking activity after users interact with these banners, including third-party tracking cookies and fingerprinting. Our analysis, based on the 33.9K websites (out of 100K) that displayed cookie banners, reveals that 47.55% of these banners use ambiguous language that blurs the distinction between cookies and other tracking technologies. In addition, we find that among these ambiguous disclosures, 43.36% mention only cookies, omitting other tracking forms. Yet these sites still show extensive tracking—a mean of 2.9 third-party tracking fingerprinting scripts. Critically, rejecting consent does not stop tracking. Among 4,581 sites that withheld third-party cookies, 18.04% still served third-party tracking fingerprinting scripts, with Google responsible for 72.1% of this activity. Finally, regional analysis reveals that Sweden and Germany exhibit the highest levels of post-rejection fingerprinting activity across all disclosure categories, demonstrating a clear gap between established GDPR frameworks and actual technical compliance.

CCS Concepts: • **Information systems** → **Web applications**; • **Security and privacy** → **Privacy protections**.

Additional Key Words and Phrases: cookie banners, online tracking, third-party tracking, fingerprinting, consent mechanisms, transparency disclosures, tracking behavior analysis, privacy compliance

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1 Introduction

Cookie consent banners are widely used to obtain user consent for online tracking, particularly following the enforcement of regulations such as the General Data Protection Regulation (GDPR) [13] and the California Consumer Privacy Act (CCPA) [10]. In practice, banners function as the primary interface through which websites disclose tracking practices and obtain consent for personal data processing activities. Such disclosures commonly involve tracking mechanisms, including cookies, advertising identifiers, or other data collection techniques. The legal validity of these consent mechanisms is closely tied to the requirements for informed consent under data protection regulations. In particular, Article 7 in conjunction with Recital 32 [27] of GDPR states: consent must be freely given, specific, informed, and unambiguous, and must be demonstrated through a clear affirmative act indicating the data subject's agreement to the processing of personal data. For consent to be considered informed and specific, the data subject must be clearly and meaningfully notified about the nature of the processing, including the types of personal data being collected and the purposes for which they will be used [26]. Building on these requirements, the EDPB Guidelines 05/2020 on Consent further clarify that consent disclosures must be sufficiently specific and informative to satisfy the conditions of informed consent under GDPR [22]. Notably, vague formulations such as "cookies and similar technologies" may not provide adequate transparency regarding the actual tracking mechanisms being deployed. Similarly, guidelines issued by the French Data Protection Authority (CNIL) emphasize that data subjects should be clearly informed about both the purpose and the nature of tracking technologies involved in data processing. These guidelines indicate that the use of generic references (e.g., "technologies" or "tools") may not be sufficient to meet the threshold for informed consent [14]. Despite these regulatory requirements, the effectiveness of cookie banners in satisfying informed consent obligations has been increasingly questioned [7].

Prior studies have shown that websites often employ deceptive patterns in consent banners to nudge users toward accepting tracking, using tactics such as pre-selected options, misleading language, and obstructive design elements that limit meaningful user choice [29, 30, 38, 41, 56]. In addition, 89% of cookie banners violated applicable legal requirements, with 61% failing legal specificity requirements by using vague justifications (e.g., "user experience enhancement") [49]. However, it remains unclear whether the disclosures in banners accurately represent the full scope of tracking activities on websites. While prior work has largely focused on banner design practices and high-level disclosure language, comparatively less attention has been given to whether cookie banners sufficiently disclose the actual tracking mechanisms deployed on websites, particularly more sophisticated forms of tracking such as browser fingerprinting. Consequently, it remains unclear whether the disclosures presented in cookie banners adequately reflect the actual scope of tracking behaviors observed in practice.

In this work, for the first time, we systematically investigate the disconnect between the tracking technologies disclosed in cookie banners and the actual tracking behaviors observed on websites. To substantiate our findings, we performed automated measurements on the Top 100K websites from the Tranco list, visiting not only the homepage but also three internal pages. We successfully collected tracking data and banner content from 33,946 websites under both consent acceptance and rejection scenarios (§ 3.1). We then leverage LLMs, which have been shown to outperform traditional privacy policy analysis tools [46], to extract disclosures and descriptions of tracking techniques from consent banners at scale, achieving 96.8% accuracy. Given their prominence in both user awareness and regulatory frameworks, we use cookies and tracking JavaScript's fingerprints as the central reference point to categorize the extracted disclosures into 4 main categories: *Transparent*, *Partially Ambiguous*, *Complete Ambiguous*, and *Not Present* disclosure (§ 3.2).

Next, we conduct extensive measurement campaigns to capture the actual tracking activity that follows user consent. This includes identifying third-party tracking cookies and fingerprinting scripts (§ 3.3). By combining natural language analysis with empirical measurement, we offer a comprehensive assessment of how tracking is communicated through banners and how these disclosures align with empirical analysis. Our main findings can be summarized as follows:

- Our results show that cookie banners overwhelmingly provide limited or incomplete disclosures. Nearly half of websites (47.55%) use ambiguous language that obscures the distinction between cookies and other tracking technologies, and within this subset, 43.36% mention only cookies while omitting stateless methods such as fingerprinting. Only a small minority explicitly acknowledge both cookies and fingerprinting, while the remaining categories—complete ambiguous and no disclosure—reflect the lowest levels of transparency (§ 4.1).
- When users interact with cookie banners, we show that accepting consent leads to large increases in both cookie-based tracking (from 20.64 to 118.96 cookies on average) and fingerprinting scripts (from 19.44 to 45.29), whereas rejecting consent produces only limited reductions, with tracking activity persisting across sites (§ 4.2).
- We confirm a disconnect between banner disclosures and actual tracking behavior, showing that many banners fail to fully inform users about the scope of tracking technologies in use. Specifically, while sites with *Partially Ambiguous* set a mean of 167.7 cookies and load 11.1 fingerprinting scripts, banners that mention *only* cookies under *Transparent Disclosure* are still accompanied by a mean of 2.14 fingerprinting scripts, despite omitting any reference to these techniques (§ 4.3).
- Rejecting consent does not effectively prevent tracking in practice. Among four disclosure groups, over 18.04% of websites that did not set third-party cookies after a user declined consent still initiated third-party tracking fingerprinting scripts, with Google accounting for 72.1% of this post-rejection tracking activity (§ 4.4).
- Across countries, websites with *Transparent Disclosure* exhibit consistent post-rejection fingerprinting activity, whereas websites with *Partially Ambiguous Disclosure* shows substantially higher cross-country variation, with Sweden (21.8%) and Germany (21.3%) recording the highest proportions (§ 4.5).

Finally, we discuss the broader technical, legal, and privacy implications of this data-practice disconnect, examining how the deployment of persistent post-rejection tracking mechanisms undermines user autonomy and compromises regulatory compliance (§ 5).

2 Related Work

Third-party web tracking refers to domains other than the visited website collecting user data across sites, browsers, and devices to support targeted advertising [5]. Although this practice is common in online advertising, it raises serious privacy concerns [3, 34, 53, 57]. In response, regulations such as the General Data Protection Regulation (GDPR)[13] and the California Consumer Privacy Act (CCPA)[10] aim to improve transparency and give users more control. Many websites now display cookie consent banners to comply with these laws, but it remains unclear how well these banners explain tracking practices or respect user choices.

With regard to the analysis of cookie banner text, prior work has shown that design choices in consent banners can strongly influence user decisions, often through the use of dark patterns.¹ Utz et al. [56] conducted large-scale experiments with over 80,000 users and showed that small interface

¹The term “dark pattern” was first introduced in 2013 to describe interface design practices that deceive or manipulate users into making choices that are not in their best interest, often through misleading layouts or selective presentation of information [9].

changes, such as banner placement, framing, and choice architecture, can significantly influence consent decisions. Graßl et al. [29] further demonstrated that design nudges like preselected options and interface obstructions reduce users' sense of control and make meaningful consent more difficult. Machuletz and Böhme [38] found that emphasizing a default "Select all" button increased consent rates, with many participants later regretting their choice. Nouwens et al. [41] analyzed common Consent Management Platform (CMP) designs and found that dark patterns and implied consent are widespread, with only 11.8% meeting their minimal GDPR requirements. Gray et al. [30] analyzed cookie banners across EU-accessible websites and emphasized the challenges in detecting and regulating dark patterns, noting that current legal standards remain insufficient. Focusing on data collection purposes, Santos et al. [49] analyzed cookie banners from a legal perspective and found that 89% violated at least one consent requirement, with 61% using vague terms that obscured the purpose of data collection. Bouhoula et al. [7] conducted an automated large-scale audit of consent interfaces, comparing stated analytic purposes with cookies set under different interaction flows. Their analysis of 97K EU websites revealed persistent discrepancies, with many sites engaging in tracking despite explicit rejection.

Overall, prior studies have shown that consent interfaces often rely on design choices that significantly influence user behavior, and have examined how cookie banners present stated purposes. However, it remains unclear whether these disclosures accurately reflect the tracking techniques employed after users interact with the banners. In this work, we bridge the gap by analyzing the relationship between the tracking techniques disclosed in cookie banners and the observed behaviors following user consent interactions. By combining large-scale analysis of banner disclosures with improved empirical measurements of third-party tracking activity, we provide the first comprehensive assessment of how stated practices align with the best possible detection of real tracking behaviors.

3 Methodology

To characterize tracking behaviors triggered by cookie banner interactions, we use *BannerClick* [45], a large-scale automated measurement tool built on top of *OpenWPM* [21] and based on Firefox. *BannerClick* is designed to detect and interact with cookie banners, enabling us to collect detailed information, including browser-set cookies, HTTP/HTTPS requests and responses, and JavaScript function calls. The tool preliminarily achieves high accuracy in automating banner interactions, with reported detection, acceptance, and rejection accuracy of 99%, 97%, and 99%, respectively [44].

Modifications to OpenWPM: A primary objective of our work is to detect fingerprinting functions executed by third-party tracking scripts loaded through HTML pages. During page load, these scripts are fetched from remote tracking vendor servers and executed in the user's browser, where they call browser-exposed JavaScript APIs, such as Canvas rendering interfaces, WebGL, and Navigator properties, to extract device and browser-specific attributes that together form a fingerprint. However, prior work has shown that OpenWPM does not reliably capture these fingerprinting scripts, missing calls on 47.8% of websites [48]. To address this limitation, we modify OpenWPM to incorporate the DuckDuckGo Tracker Radar dataset [19] as a reference for known tracking and fingerprinting techniques. Tracker Radar is widely used for identifying tracking-related behaviors [35, 39, 51].

Fingerprinting Instrumentation Evaluation: We evaluate our fingerprinting instrumentation through a comparative analysis between the baseline *OpenWPM* instrumentation and our modified framework. While the baseline OpenWPM configuration primarily instruments classical fingerprinting surfaces such as Web Audio APIs (*AudioContext* and *OfflineAudioContext*) and a limited set of browser-exposed environment properties (e.g., screen-related attributes such as *pixelDepth* and *colorDepth*), our modified instrumentation expands the set of monitored fingerprinting surfaces

to include WebGL rendering APIs, hardware and device enumeration interfaces, media capability probing, sensor APIs, font enumeration mechanisms, navigator attributes, timing interfaces, and browser environment properties. To quantify the impact of these modifications, we independently crawled the Tranco Top 5K domains using both configurations under identical experimental conditions. Both crawls were conducted from the same vantage point (Germany), using identical browser settings, interaction logic, and consent actions (accept mode).

As shown in Table 1, we compare fingerprinting instrumentation coverage between the baseline OpenWPM and our modified OpenWPM implementation across 11 categories of fingerprinting techniques, ranging from conventional fingerprinting surfaces (e.g., *Navigator objects* and *display properties*) to more advanced mechanisms such as WebGL, sensor-based, keystroke, and media I/O fingerprinting. We highlight the differences in instrumentation support between the baseline and extended framework, as well as representative browser-exposed JavaScript methods and properties introduced to improve coverage for each fingerprinting category.

Table 1. Comparison of instrumentation support between baseline and modified OpenWPM across fingerprinting categories. *Partial* indicates categories where baseline OpenWPM provided limited instrumentation support. *Absent* denotes fingerprinting categories for which baseline OpenWPM did not provide instrumentation support. *Extended* refers to categories where existing instrumentation was expanded through additional API methods, or properties, whereas *Added* denotes fingerprinting categories newly introduced into our modified framework.

Category	Baseline	Modified	Extended API Methods/ Properties	Extended Fingerprinting Relevance
Screen/Display Fingerprinting	Partial	Extended	<code>devicePixelRatio</code> , <code>screen.orientation</code> , <code>matchMedia</code>	Captures DPI scaling factor, screen orientation state and CSS media query signals
Audio Fingerprinting	Partial	Extended	<code>AudioBuffer</code> , <code>AudioWorkletNode</code>	Captures waveform, buffer-level extraction and worklet DSP execution
WebRTC Fingerprinting	Partial	Extended	<code>RTCPeerConnectionIceEvent</code>	Derives network signaling behavior, peer connection state dynamics and communication telemetry signals
Performance and Time Fingerprinting	Absent	Added	<code>Performance.now</code> , <code>requestAnimationFrame</code> , <code>Performance.getEntries</code>	Derives execution timing jitter, rendering pipeline timing and resource loading traces
Media I/O Fingerprinting	Absent	Added	<code>MediaDevices.enumerateDevices</code> , <code>MediaSource.isTypeSupported</code> , <code>speechSynthesis.getVoices</code>	Exposes hardware media enumeration, codec support signals and speech synthesis voice profiles
DOM Layout fingerprinting	Absent	Added	<code>getBoundingClientRect</code> , <code>offsetWidth</code> , <code>scrollHeight</code>	Derives DOM layout computation differences, element geometry variance and rendering overflow characteristics
Font Fingerprinting	Absent	Added	<code>document.fonts.load</code> , <code>document.fonts.check</code> , <code>FontFaceSet.check</code>	Exposes font availability probing, font loading behavior and font set validation signals
WebGL Fingerprinting	Absent	Added	<code>getParameter</code> , <code>getSupportedExtensions</code> , <code>getShaderPrecisionFormat</code>	Exposes GPU renderer parameters, extension surface and shader precision queries
Keystroke Fingerprinting	Absent	Added	<code>KeyboardEvent.key</code> , <code>KeyboardEvent.location</code> , <code>KeyboardEvent.timeStamp</code>	Captures keystroke timing patterns, key identity mapping and keyboard layout position signals
Sensor-based Fingerprinting	Absent	Added	<code>DeviceMotionEvent.acceleration</code> , <code>DeviceOrientationEvent.alpha</code> , <code>DeviceOrientationEvent.rotationRate</code>	Derives motion sensor variability, spatial orientation vectors and rotation rate signatures
Cross-Context Communication Fingerprinting	Absent	Added	<code>BroadcastChannel.postMessage</code> , <code>Worker</code> , <code>MessageChannel</code>	Exposes parallel execution behavior, cross-tab communication behavior and message passing coordination

Building on the instrumentation extensions described above, we analyse baseline and modified OpenWPM to quantify the additional fingerprinting coverage introduced by our extensions. The

Table 2. Comparison of fingerprinting detection coverage between baseline and modified OpenWPM across 1,945 banner-enabled websites.

Metric	Baseline	Modified	Gain
% Websites with ≥ 1 fingerprinting API detected	53%	53.62%	0.62%
% Websites with Performance & Time fingerprinting	0	36.91%	36.91%
% Websites with Media I/O fingerprinting	0	18.14%	18.14%
% Websites with DOM Layout fingerprinting	0	0.56%	0.56%
% Websites with Keystroke fingerprinting	0	1.07%	1.07%
% Websites with Font fingerprinting	0	0.05%	0.05%
% Websites with Screen/Display fingerprinting	0	18.09%	18.09%
Median unique fingerprinting methods/properties per website	3	4	$\Delta = 1$

evaluation was conducted on the Tranco Top 5K domains, resulting in 1,945 banner-enabled websites (see Table 2). Based on the coverage transitions in our modified version, we quantify improvements across two dimensions. For categories with existing baseline support that were further extended (Partial $\rightarrow$ Extended), we measure the additional fingerprinting methods and properties intercepted by the modified framework. As shown in Table 2, the median number of unique fingerprinting methods/properties increased from 3 to 4 per website, yielding a median increase of one additional fingerprinting element. The broader instrumentation also increased overall detection coverage, with the proportion of websites exhibiting at least one fingerprinting API increasing from 53.0% (1,031/1,945) to 53.62% (1,043/1,945).

3.1 Data Collection

We selected the Top 100K domains from the Tranco list [33], which provides a stable, reproducible ranking of popular websites. For each domain, we open the website’s landing page only once, in a fresh browser profile and periodically check for the presence of a consent banner at 10-second intervals, until a banner is detected. This process is bounded by a maximum duration of 90 seconds to accommodate delayed banner rendering while ensuring a fixed upper limit on observation time, and is applied uniformly across websites to capture initial tracking behavior prior to the appearance of consent banners. We then accept the cookie banner, consent to all tracking, wait 10 seconds, and reload the page. Since crawling only homepages is no longer considered best practice, as inner pages can exhibit markedly different characteristics, including distinct privacy behaviors [2, 15, 55], we also visit three *internal pages* for each website. To do so, we select a link by crawling `<a>` elements and check whether it is a potential inner page. We verify that the inner-page link must begin with the landing page’s fully qualified domain name. We further access the inner pages and perform random mouse movements and automatic scrolling from top to bottom. This emulates typical user behavior and provides a more realistic view of tracking.

We applied the same procedure for the reject scenario, where we explicitly rejected all tracking through the banner. During this process, we extracted the cookie banner text using BannerClick’s functionality by locating the banner element in the DOM (including iframes and Shadow DOM) and retrieving its visible text via the “.text” attribute.² The crawl was conducted using BannerClick’s

²Note that websites are often available in languages other than English. Across the 33,946 collected banners, approximately 70% were originally presented in English, while the remaining banners spanned a total of 59 distinct languages. Since many languages appeared only rarely in the dataset, we focused our translation validation on languages representing at least 1% of the collected banners, resulting in 10 prominent languages selected for evaluation. To assess the reliability of the translation pipeline, we randomly sampled 10 banners from each selected language and manually verified the translated

default settings in April 2026, from a single machine hosted in Europe. Using this crawling pipeline, we successfully accessed 83,175 websites from the Tranco Top 100K list, among which BannerClick extracted 33,946 consent banners for downstream analysis.³ For the remaining websites, we observed three outcomes: (i) access was blocked by CAPTCHA, (ii) the crawler was redirected to anti-automation pages, or (iii) the website did not display a consent banner. For the extracted banners, we successfully executed both accept and reject interaction scenarios and stored detailed information together with corresponding banner screenshots for subsequent analysis. Additionally, we also crawled Tranco Top 5K domains from seven geographical locations (Germany, Brazil, South Africa, Sweden, US-East, US-West, and Australia) using BannerClick to analyze geographical disparities in disclosure practices and post-rejection tracking behavior (see § 4.5 for details).

3.2 Identifying Tracking Techniques Disclosed in Cookie Banners

Beyond static keyword matching or rule-based approaches, we leverage LLMs to extract tracking techniques disclosed in cookie banners. Specifically, we use OpenAI’s GPT-4.1 (gpt-4.1-2025-04-14), selected for its state-of-the-art performance in natural language understanding, structured information extraction, and privacy-related reasoning [42]. We adopt a zero-shot prompting approach, which enables the model to generalize across diverse banner texts without requiring labeled examples or task-specific fine-tuning. This approach avoids the cost and potential bias of training data collection and has proven successful in recent research across software engineering, policy analysis, and empirical studies involving large language models [12, 31].

Note that we refined our zero-shot prompts using the “Prompt Professor” assistant [11], a framework designed to improve prompt contextuality and performance by clearly specifying task details, instructions, and rules defined. As part of this refinement, we appended an additional extraction field, “explanation”, which requires the model to provide a concise justification for the selected categories. This supports systematic inspection of model reasoning and helps identify potential sources of misclassification in the extracted tracking attributes.

Given cookies are the most widely recognized method for identifying users [1, 21, 47], and are often the primary focus of both user attention and regulatory consent mechanisms, we treat cookie-related terminology as a stable reference signal within disclosure text. To avoid introducing additional semantic assumptions, we rely solely on Boolean indicators derived directly from the banner text. Our classification proceeds in two steps: we first determine the presence of vague disclosure terms, and then incorporate explicit signals related to cookies and stateless tracking technologies. Our prompt (see Appendix B), therefore, guides the model to classify banner text into the following three different classes.

- **Cookies-related:** If the banner text has terms about stateful behavior, e.g., “cookies”, “session storage”, LLM marks the class as [C].
- **Fingerprinting-related:** If the banner text has terms related to stateless tracking, such as “JavaScript”, “geolocation”, etc., LLM marks the class as [F].
- **Vague:** If the banner text contains generic expressions such as “tracking technologies” or “similar technologies”, they are assigned [V] in the absence of explicit contextual grounding in concrete tracking mechanisms.

Please note that a banner may belong to all three classes and may also not belong to any class, e.g., if the banner has no disclosure of tracking.

output with native speakers. This evaluation confirmed that the translations introduced no substantial semantic errors or changes in meaning, providing confidence in the correctness of the translated banner text used for further analysis.

³Note that the BannerClick tool was recently reevaluated in [44], achieving detection and interaction accuracies >97%.

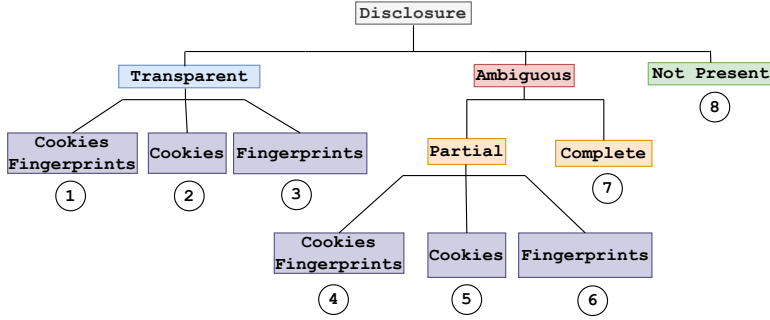


Fig. 1. Disclosure taxonomy categorizing cookie-banner text by transparency, ambiguity, and specificity of cookies and fingerprinting disclosures.

Validation: Using BannerClick within our web measurement pipeline, we successfully detect and collect 33,951 candidate cookie banners. We then apply our LLM-based extraction pipeline over all collected banners and analyze cases where the model failed to produce valid responses. This failure analysis reveals 15 edge cases requiring further inspection. Among these, we identify five false positives where non-consent interface elements had been incorrectly detected as cookie banners, reducing the final count to 33,946 valid banners. The remaining cases correspond to intermediate pipeline failures, including 7 Chinese-language banners where the *LibreTranslate*-based translation stage failed before downstream LLM analysis, which we subsequently resolved by translating the extracted banner text using *Google Translate*, and 3 websites where BannerClick failed to recover banner content due to parsing limitations, which were resolved through manual inspection. To further verify the reliability of the cookie-banner text extraction process, we manually inspected a random sample of 500 banners from the final set of 33,946 collected banners. For each sample, we compared the text extracted by the BannerClick-based pipeline against the cookie banner visibly displayed on the corresponding website. The inspection showed 498 banners correctly extracted, corresponding to 99.6% extraction accuracy. In the remaining cases, the cookie banner was successfully detected; however, the extracted content primarily contained button text instead of banner text. These results demonstrate the high reliability and accuracy of the banner text extraction process.

Evaluation: To evaluate the accuracy of our LLM-based extraction approach, which is applied across all 33,946 websites, we randomly sample 1,000 cookie banners from the complete dataset for manual evaluation. One of the authors annotates all 1,000 banners, while the other two authors each independently annotate a stratified subset of 300 banners, sampled as 100 from the top, 100 from the middle, and 100 from the bottom of the sampled results, yielding 300 banners jointly labeled by all three annotators.

Each banner is annotated into three categories: Cookies-related (C), Fingerprinting-related (F), or Vague (V), based on the semantic content of the banner text. We assign binary indicators for each category based on whether the corresponding tracking techniques were present, which in turn yields a categorical classification from these Boolean values (see Table 3). For example, a banner stating, “*We use cookies and identifiers such as IP address...*” explicitly refers to cookies and additional tracking identifiers, resulting in $C = \text{True}$, $F = \text{True}$, and $V = \text{False}$, thereby falling under the Transparent Disclosure category.

Based on this annotation procedure, inter-annotator agreement is computed on the 300 jointly labeled banners to assess labeling consistency. We observe strong agreement of 98%, indicating reliable classification across annotators. Manual inspection further showed that the LLM-based

Table 3. Disclosure classification based on terms related to cookies (C), Fingerprinting (F), and vague (V).

Case	Logic	Example Scenario
1	$\neg V \wedge C \wedge F$	"We use cookies and identifiers like IP address..."
2	$\neg V \wedge C \wedge \neg F$	"We use cookies for analytics..."
3	$\neg V \wedge \neg C \wedge F$	"We use device identifiers..."
4	$V \wedge C \wedge F$	"We use cookies and similar techniques..." + "track your IP address..."
5	$V \wedge C \wedge \neg F$	"We collect information..." + "cookies..."
6	$V \wedge \neg C \wedge F$	"tracking techniques..." + "geolocation is tracked"
7	$V \wedge \neg C \wedge \neg F$	"We process your information..."
8	$\neg V \wedge \neg C \wedge \neg F$	No disclosure text

extraction incorrectly identified the disclosed tracking techniques in 32 of the 1,000 sampled banners (3.2%), resulting in an overall accuracy of 96.8%. We provide the details about cases where LLM extraction failed in Section A. Our approach performs well primarily for two key reasons.

First, the relatively small input context (only the banner text), allows the LLM to focus more effectively on the relevant entities and make more accurate decisions. Second, our prompt instructs the LLM to provide a brief explanation for each decision, thereby guiding the LLM toward the correct direction and allowing us to re-evaluate the prompt's effectiveness during manual validation. Third, we introduce explicit instructions for handling cases in which tracking-related terms may appear linguistically vague but depend on contextual grounding. In particular, the prompt requires the model to prioritize surrounding contextual evidence over isolated keywords when deciding whether a term should be classified as vague or grounded. This ensures that classification relies on concrete contextual evidence of tracking (or its absence), rather than surface-level lexical patterns, improving consistency in distinguishing vague disclosures from contextually grounded ones. Although semantic granularity remains a critical concern in LLM decision-making, this design enables the model to accurately extract relevant information while minimizing the influence of irrelevant context.

3.3 Tracking Resource Detection

We identify tracking behaviors on websites by analyzing cookies and fingerprinting scripts. To determine whether a resource is third-party, we rely on the Public Suffix List [40], comparing the registrable domain of each request or cookie against the visited domain. If the two domains differ, the resource is classified as third-party. To identify third-party tracking cookies, we utilize the *justdomains* list [32], which consolidates domains from multiple tracking blocklists. If the host of a third-party cookie matches a domain in the *justdomains* list, we classify it as a tracking cookie. This domain-based approach offers high coverage and is consistent with methods used in prior tracking studies [28, 45, 52]. To detect third-party tracking-related fingerprinting, we use a rule-based approach grounded in widely adopted filter lists. Specifically, we apply *EasyList* [24], *EasyPrivacy* [25], and *Adservers* [37] blocklists that are routinely used by browser extensions to filter advertising and tracking content. We match all third-party scripts against these lists using the *Adblockparser* [50], a parser that implements the filtering logic of Adblock Plus. We then cross-reference the domains serving these scripts with the *Disconnect* tracker list [16], which categorizes third-party services based on their tracking behavior, including fingerprinting. Unlike general-purpose blocklists such as *EasyList*, which focus on ad-related requests, the *Disconnect* list includes a specific fingerprinting category, allowing us to identify scripts associated with entities known to engage in fingerprint-based tracking. The data analyzed corresponds to measurements collected as of April 2026.

3.4 Limitations

Our analysis relies on widely adopted blocklists, viz., *EasyList*, *EasyPrivacy*, *Adservers*, and *Disconnect*, to identify third-party trackers and fingerprinting activity. These lists are commonly used in both academic research [21, 54] and privacy tools deployed in practice. While these lists provide a solid foundation for measuring observable tracking behaviors, they may not detect all instances of data collection, particularly by previously unknown or evasive actors. We therefore interpret our measurements as a lower bound on the actual prevalence of tracking activities. Similarly, the accuracy of our findings also depends on the reliability of the LLM-based extraction pipeline used to interpret banner text. In this context, while our LLM-based approach achieves high accuracy (96.8%), it still exhibits a small number of residual errors arising from contextual interpretation of banner text. These errors occur primarily where the model must distinguish whether a given phrase reflects an explicit data disclosure practice or instead conveys general information. They are largely driven by limitations in capturing implicit contextual grounding, particularly when tracking-related terms are weakly supported, appear outside the local context, or involve subtle distinctions between vague and stateless tracking references. Although limited in number, these cases indicate that the model is sensitive to contextual ambiguity in distinguishing explicit disclosure statements from indirectly expressed tracking references. We also note that our crawling setup does not explicitly account for whether websites detect automation. Bot detection mechanisms may modify page content or tracking behavior, potentially affecting observed measurements. Since our study relies on a standardized crawling configuration with minimal user interaction, we do not control for these effects, which should be considered when interpreting the results.

4 Results

In this section, we analyze the disconnect between tracking disclosures in cookie banners and observed tracking behaviors, including the distribution of tracking technique disclosures, tracking behavior across disclosure categories, and tracking behavior after consent rejection without cookies.

4.1 Distribution of Tracking Technique Disclosures

Table 4. Distribution of 33,946 websites (with banners) by tracking disclosure.

Category/Sub Category	Count / %
Transparent Disclosure	17,257 / 50.83%
Cookies & Fingerprinting	478 / 1.4%
Cookies	16,774 / 49.41%
Fingerprinting	5 / 0.01%
Ambiguous Disclosure	16,142 / 47.55%
Partially Ambiguous	15,818 / 46.59%
Cookies & Fingerprinting	8,468 / 24.94%
Cookies	7,000 / 20.62%
Fingerprinting	350 / 1.03%
Complete Ambiguous	324 / 0.95%
Empty / No Disclosure	547 / 1.61%
Total	33,946 / 100%

We applied our taxonomy to 33,946 websites whose cookie banners were processed successfully using our LLM-based extraction method. As shown in Table 4, the most frequent category is *Transparent Disclosure – Cookie*, found on 16,774 websites (49.41%). These banners reference only cookies, omitting other tracking technologies, and reflect a narrow focus on stateful methods. This raises concerns about whether users are adequately informed, particularly in contexts where stateless tracking is known to be prevalent [8, 21, 36]. The second most common category is *Ambiguous disclosure*, observed on 16,142 websites (47.55%). These banners mention cookies but also use imprecise terms such as “technologies” or “tools,” blurring the line between explicit and ambiguous communication. This use of generalized language may reduce users’ ability to understand what types of tracking are actually employed. Interestingly, 478 websites explicitly

mention cookies along with fingerprinting, indicating a more comprehensive acknowledgment of the tracking mechanisms in use. In addition, we observed 324 websites (0.95%) that fall into the *Complete Ambiguous Disclosure* category, where tracking is referenced only through vague or ambiguous terms without naming any specific mechanism. Finally, 547 websites (1.61%) do not mention anything related to tracking techniques in their banner at all. Although both categories are relatively uncommon, the absence of any tracking-related disclosure (1.61%) represents the lowest level of transparency observed in our dataset.

Key takeaway: Our taxonomy shows that most websites disclose tracking incompletely. Nearly half (47.55%) use ambiguous language that fails to clearly distinguish cookies from other tracking technologies, and 43.36% of these disclosures mention only cookies while omitting stateless techniques such as fingerprinting. Only 478 sites explicitly disclose both cookies and fingerprinting. The least transparent categories, complete ambiguity (0.95%) and no disclosure (1.61%), are relatively rare. Overall, most banners provide disclosures that are broad, incomplete, or silent about the full range of tracking mechanisms they employ.

4.2 Before and After Interaction

In this section, we analyze tracking behavior before and after user interaction across 33,946 websites (Figure 2). For third-party tracking cookies, the mean count increases from 20.64 to 118.96 after consent acceptance, whereas rejecting consent results in a smaller increase from 15.74 to 23.56. For fingerprinting scripts, we observe a similar pattern: the mean rises from 19.44 to 45.29 after acceptance, while rejection increases fingerprinting activity only from 18.68 to 29.83. Notably, fingerprinting activity persists even on sites that set no third-party cookies. In this subset, fingerprinting scripts increase from a mean of 4.54 to 5.49 after acceptance, and from 2.77 to 5.71 after rejection.

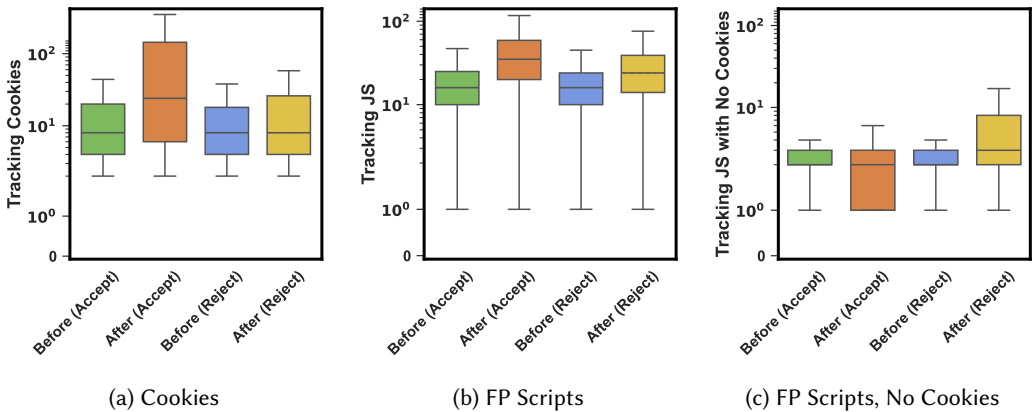


Fig. 2. Number of third-party tracking cookies, fingerprinting tracking scripts observed before and after user interaction across 33,946 websites.

4.3 Tracking Behavior After Acceptance

To assess the alignment between what websites disclose and their actual tracking behavior upon accepting the banner, we analyzed the number of (1) third-party tracking cookies and (2) fingerprinting tracking scripts present on each site in Figure 3. Websites categorized under *Partially Ambiguous*

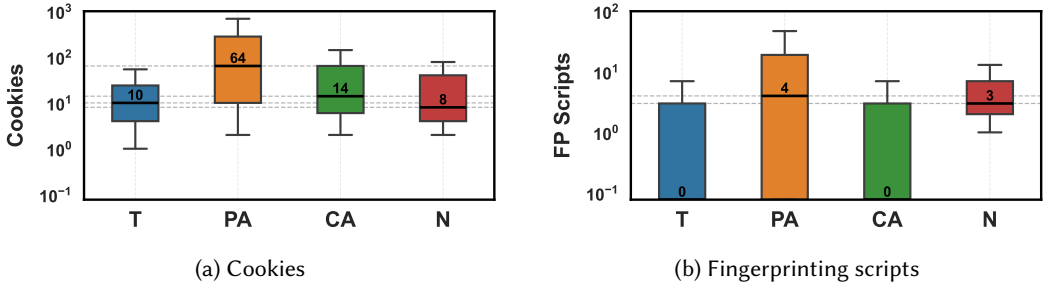


Fig. 3. Tracking cookies and fingerprinting scripts observed after accept, grouped by disclosure type across 33,946 websites (T: Transparent, PA: Partially Ambiguous, CA: Completely Ambiguous, N: No Disclosure).

disclosure exhibit the highest levels of tracking across both measured metrics. These sites set a mean of 167.6 third-party tracking cookies and embed a mean of 11.1 fingerprinting-related scripts.

The next highest category is *Completely Ambiguous* disclosure, with a mean of 123.6 third-party tracking cookies and a mean of 3.8 fingerprinting-related scripts. In comparison, websites with *Transparent* disclosure set a mean of only 34.9 third-party tracking cookies and embed 2.5 fingerprinting-related scripts. This trend suggests that more comprehensive banner disclosures often correlate with more extensive tracking practices. Interestingly, websites categorized under *No Disclosure* still set a mean of 57.7 third-party tracking cookies and a mean of 6.9 fingerprinting-related scripts. These findings indicate that less informative, ambiguous, or entirely absent disclosures are often associated with more extensive and less transparent tracking behavior.

In addition, when we look at the sub-category, we find that websites in the *Transparent - Cookies* disclosure still engage in measurable levels of fingerprinting scripts, despite their banners omitting any mention of these techniques. Specifically, the mean number of fingerprinting scripts is 2.13 in this group. This indicates that banners referencing only cookies substantially underrepresent the full scope of tracking technologies.

Key takeaway: *Partially Ambiguous* websites exhibit the highest tracking intensity. However, banners that mention *only* cookies still accompany a mean of 2.13 fingerprinting scripts, despite omitting any reference to these techniques, highlighting a persistent gap between disclosed and actual tracking practices. Notably, *No Disclosure* websites still show substantial tracking, with a mean of 57.7 third-party cookies and 6.9 fingerprinting scripts, suggesting that minimal or absent disclosures often coincide with higher, less transparent tracking.

4.4 Tracking Behavior After Rejection

To assess whether websites engage in tracking even after users reject consent, we analyzed 25,380 sites (out of 33,946 sites) that did not set any third-party cookies following a rejection interaction. We then measured the proportion of these sites that initiated third-party tracking fingerprinting scripts despite the absence of tracking cookies. As shown in Figure 4, over 18.04% of sites across all disclosure categories issued third-party tracking fingerprinting scripts despite the absence of tracking cookies (4,581 out of 25,380 sites). This behavior was most prevalent among sites with *Partially Ambiguous* disclosure (24.4%) and *Completely Ambiguous* disclosure (17%). For 14% of *Transparent* disclosure sites, when we focus on sites that mention only *cookie*, we find that 13.34% still deploy fingerprinting scripts. Additionally, 6.6% of sites with *No Disclosures* were observed to initiate third-party tracking fingerprinting scripts despite providing no prior indication of either cookie-based or fingerprinting-based tracking in banner text. This suggests that the absence of

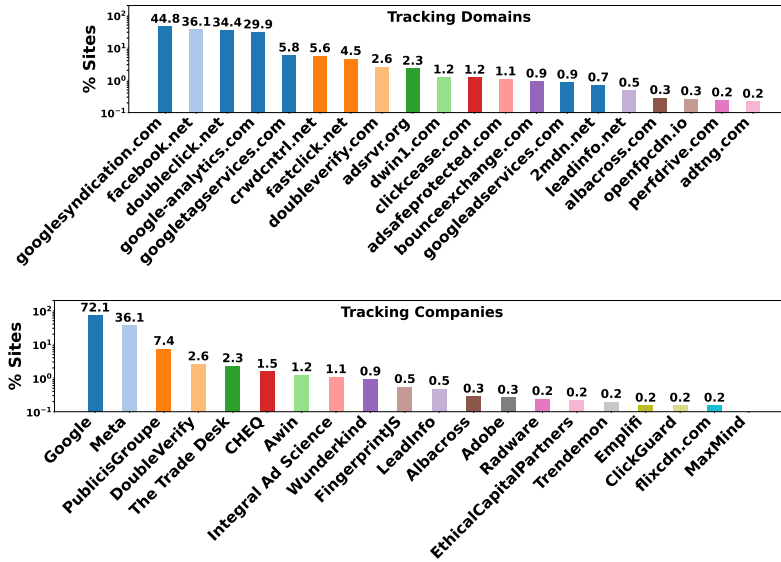


Fig. 5. Percentage of 4,581 websites triggering fingerprinting tracking scripts despite not setting cookies after consent rejection. The top panel shows the 20 most tracking domains; the bottom panel aggregates them by company, matched via the *Disconnect* list [17]. Bars are color-coded by company across both plots.

disclosure does not necessarily correspond to the absence of tracking-related behavior. These findings demonstrate that rejecting cookies does not necessarily prevent other forms of third-party communication commonly associated with tracking.

Recent studies have analyzed cookie banners and tracking practices primarily through the presence or absence of third-party cookies [6, 45]. However, our findings show that such cookie-centric approaches may significantly underestimate the true extent of web tracking. Even in the absence of tracking cookies, over 18.04% of websites continue to engage in third-party communication and load fingerprinting scripts. This underscores the need for privacy measurement methodologies that incorporate behavioral analysis and go beyond static cookie-based indicators.

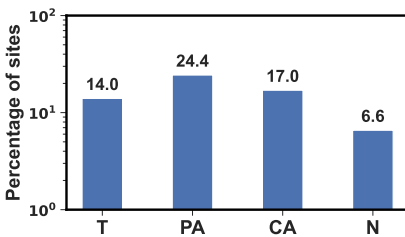


Fig. 4. Sites that loaded fingerprinting scripts despite not setting any tracking cookies after consent rejection (T: Transparent, PA: Partially Ambiguous, CA: Completely Ambiguous, N: No Disclosure).

At the company level, Google accounts for 72.1% of the sites, far ahead of Meta (36.1%)

To identify the actors involved in post-rejection tracking, we analyzed the third-party tracking fingerprinting scripts sent by these websites that did not set any third-party cookies after users rejected consent. Figure 5 presents the 20 most contacted tracking domains (top) and their aggregated ownership by company (bottom), matched using the *Disconnect* tracking protection list [17]. Google services dominate post-rejection tracking. googlesyndication.com alone appears on 44.8% of sites, beside this facebook.net appears on 36.1% of sites followed by other Google-owned domains such as doubleclick.net (34.4%) and google-analytics.com (29.9%).

Key takeaway: Even when third-party cookies are withheld after rejection, other tracking vectors remain active: 18.04% of sites continue deploying fingerprinting tracking scripts. In addition, tracking infrastructure remains active after consent rejection, often via requests to embedded scripts from major platforms, even in the absence of cookies.

4.5 Impact of Geographical Location

In this section, we analyze tracking behavior across seven geographical locations after banner rejection, grouped by disclosure type. Using BannerClick, we independently crawl the Tranco Top 5K domains from Germany, Brazil, South Africa, Sweden, US-East, US-West, and Australia to study regional variation in disclosure practices and post-rejection fingerprinting behavior.

Firstly, we observe that banner occurrence varied significantly by region. Sweden (39.56%) and Germany (38.84%) exhibited the highest proportion of detected cookie banners, while occurrence in the remaining five regions hovered closely between 24% and 28%. Among websites displaying a banner, consent rejection was highly effective at preventing third-party cookies across all jurisdictions; the proportion of cookie-free websites remained remarkably uniform, ranging from 73.01% in Australia to 76.39% in Sweden.

Restricting to this subset, we further analyze how these websites are distributed across disclosure categories and geographical locations. As shown in Figure 6, in Germany, nearly half of these websites fall under *Transparent* disclosure (48.7%, 712/1,463). A comparable fraction exhibits *Partially Ambiguous* disclosure (47.7%, 697/1,463), while a smaller share corresponds to *Completely Ambiguous* disclosure (2%) and *No Disclosure* (1.4%). While the absence of cookies after consent rejection may suggest stronger compliance behavior, it remains important to examine the extent to which these websites continue to engage in alternative forms of tracking, particularly fingerprinting-related techniques. To examine this further, we analyze its prevalence in a stratified manner across disclosure categories and geographical regions to assess how it varies with banner-level disclosure practices. Germany exhibits a nearly equal distribution between *Transparent* disclosure (48.7%) and *Partially Ambiguous* disclosure (47.7%) among websites that do not set cookies after rejection. However, the prevalence of fingerprinting-related tracking differs substantially across these categories. Among websites with *Transparent* disclosure, 9.8% continue to employ fingerprinting-related tracking, whereas this proportion increases to 21.3% for websites belonging to the *Partially Ambiguous* disclosure category.

Across countries, websites in the *Transparent* disclosure group exhibit relatively stable levels of post-rejection fingerprinting activity across regions. In contrast, the *Partially Ambiguous* disclosure category shows substantially higher variation, with Sweden (21.8%) and Germany (21.3%) recording the highest proportions. Comparatively, websites with *Complete Ambiguous* disclosure and *No Disclosure* exhibit lower and less consistent patterns of fingerprinting activity across regions, without a clear cross-country trend.

Key takeaway: Sweden and Germany consistently exhibit elevated levels of post-rejection fingerprinting activity across all disclosure categories. This persistent pattern suggests that differences in banner-level disclosure practices alone do not fully explain post-rejection tracking behavior, raising broader concerns about the effectiveness of current transparency and consent mechanisms, particularly in jurisdictions operating under established regulatory frameworks such as GDPR.

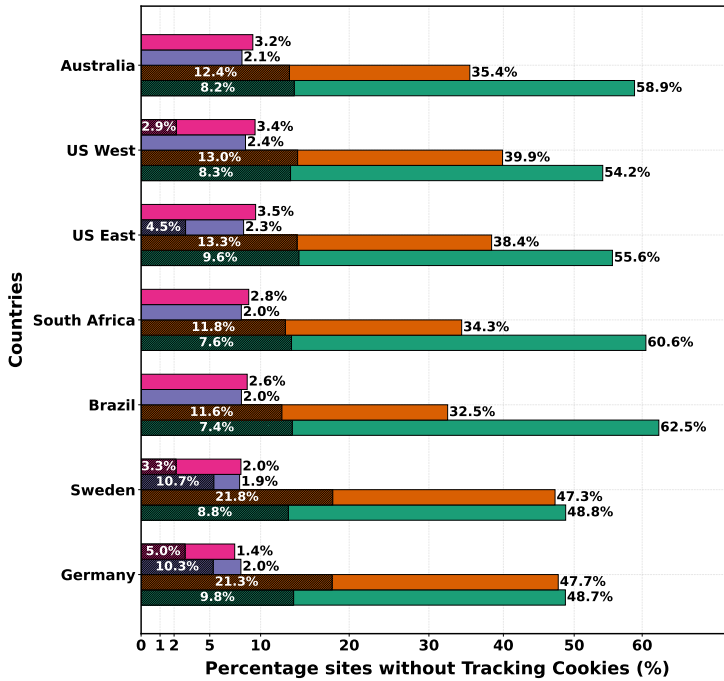


Fig. 6. Share of Tranco top 5K domains that set no cookies after consent rejection, stratified by disclosure category and region; darker hatched segments indicate the proportion within each category that loaded fingerprinting scripts. Different disclosure types are shown in different colors: Transparent (●), Partially Ambiguous (●), Completely Ambiguous (●), No Disclosure (●).

5 Discussion

Our results reveal a clear disconnect between the behavior websites declare in their consent banners and their actual data practices. We consistently observe the continued deployment of persistent tracking mechanisms after explicit rejection, including fingerprinting techniques capable of collecting device- and browser-specific information. Such practices raise important concerns regarding whether current consent interfaces adequately and transparently communicate the nature and scope of personal data processing that continues to occur in practice. Some users may even assume that simply blocking or deleting cookies in their browsers is sufficient to protect their privacy, unaware that more sophisticated tracking techniques can still compromise their data autonomy. Moreover, such practices may violate privacy regulations, including the GDPR. While the GDPR outlines key requirements for valid consent—specifically that it must be freely given, specific, informed, and unambiguous (Article 4(11) and 7 GDPR) [13]—it does not mandate the technical implementation of these standards. Instead, their interpretation has been guided by non-binding guidelines such as those issued by the European Data Protection Board (EDPB) [23] and national Data Protection Authorities (DPAs) [4]. Notably, the EDPB’s Guidelines 05/2020 on Consent emphasize that consent must be both specific and informed, explicitly stating that vague phrasing such as “cookies and similar technologies” is insufficient [22]. Similarly, the French DPA (CNIL) requires that users be clearly informed about the purposes, nature of tracking technologies, and discourages generic references (e.g., “technologies” or “tools”) which may not satisfy the threshold for informed consent [14]. Building on this gap, we emphasize that cookie banners should explicitly state the underlying tracking mechanisms as the principal basis for describing

the different modes of data collection involved. Rather than relying solely on abstract references such as “tracking technologies,” disclosures should be grounded in explicit contextual evidence that links high-level categories to their concrete technical implementations. While explicitly listing every data element involved in tracking may not be practical in real-world deployments, we instead emphasize the importance of highlighting representative and concrete contextual evidence that captures the underlying tracking behavior. Rather than attempting exhaustive disclosure, such selective contextual grounding can provide users with clearer and more meaningful insights into the broader nature of data collection practices through a limited set of illustrative examples. This approach offers a more balanced formulation that preserves legal conciseness while improving transparency and aligning better with the requirements of informed consent under data protection regulations.

6 Conclusion

In this paper, we conducted a large-scale analysis of cookie banner disclosures and the tracking behaviors they accompanied. By combining language model-based extraction with empirical web measurements, we revealed a persistent gap between declared tracking practices and actual data collection. Our findings showed that websites frequently continue tracking-related activities despite rejecting consent, and that fingerprinting-related mechanisms remain prevalent even among websites that avoid setting cookies after rejection. Furthermore, we observed that websites with partially ambiguous disclosures exhibit substantially higher prevalence of fingerprinting-related tracking compared to websites providing transparent disclosures, indicating that banner transparency alone is not a reliable indicator of privacy-preserving behavior. These findings raise broader concerns about the reliability of current consent mechanisms and highlight the need for stronger transparency and enforcement practices on the Web.

Availability: To support open science policy, we shared an open replication package containing our crawler, raw data, analysis scripts, and visualizations: <https://doi.org/10.5281/zenodo.21708043>

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Ethics and Privacy Statement

Our study investigates the relationship between tracking behavior and consent banner disclosures, focusing exclusively on publicly accessible websites. All interactions were conducted strictly for research purposes, with no intention to disrupt, exploit, or interfere with website functionality. We collected data using automated browsing configured to replicate typical user behavior through *OpenWPM*, without logging into any accounts, submitting forms, or bypassing access controls. We follow established ethical principles for network and web measurement as outlined by Partridge and Allman [43] and Kenneally and Dittrich [18], and adhere to best practices described by Durumeric et al. [20]. We do not store or analyze any sensitive user data, and our analysis is limited to observable tracking behaviors as experienced by a standard browser user.

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A Evaluation of LLM Classification

In Section 3.2, we show that the LLM-based extraction incorrectly identified the disclosed tracking techniques in 32 of the 1,000 sampled banners (3.2%), resulting in an overall accuracy of 96.8%. In particular, there were 10 cases in which the relationship between vague disclosure terminology and the corresponding tracking evidence was indirect or embedded within broader contextual statements. For instance, statements such as—*collect other information (e.g., device characteristics, supported technologies)* required the model to infer that the referenced contextual evidence still constituted vague disclosure rather than explicit tracking disclosure; the LLM underperformed in a small number of such structurally implicit cases. We further observe seven cases where the model failed to classify generic terms such as “personal data,” “data,” or “device characteristics” as vague disclosures when these terms appeared without sufficient contextual grounding to a specific tracking mechanism. In 5 additional cases, relevant contextual information was present, but appeared outside the immediate surrounding text of the vague term, limiting the model’s ability to correctly associate it with the corresponding evidence. The remaining failures involve narrower semantic interpretation challenges. In four cases, vague tracking terminology was clearly clarified by the surrounding context, but the model still classified the disclosure as vague. Similarly, four cases involved generalized stateless-tracking terminology, where slight semantic variations led to missed fingerprinting-related disclosures. For example, banners referring to “*location*” rather than the more explicit “*geolocation*” led to misclassification in some rare and isolated instances. Finally, in 2 cases, the model incorrectly inferred fingerprinting-related tracking despite the absence of explicit stateless-tracking evidence in the banner text. Based on this sample, we consider the observed error rate sufficiently low to proceed with the large-scale analysis presented in the next section.

B LLM Prompt Template

The following prompt was used to extract structured tracking technique information from cookie banner texts in a zero-shot setting.

```
You are an expert in web tracking techniques and privacy-policy analysis.
### TASK – Extract Tracking Details
Carefully analyze the provided cookie-banner text and extract the following fields:
---
## 1. "tracking_techniques"
- List all tracking mechanisms or categories explicitly mentioned.
- Normalize all terms to lowercase.
- Include a term only if it meets ALL conditions:
  1. It appears explicitly in the text.
  2. It is described as being used or potentially used.
  3. It is *not* mentioned only in a negated/denial context.
### Allowed Technique Categories (Include only when explicitly named)
- Stateful Tracking (Targeting Cookies): "cookies", "local storage", "session storage"
- Fingerprinting (Stateless Method): "device scanning", "device identifiers", "precise
↳ geolocation", "ip address"
```

```

- Vague Terms: "tracking technologies", "similar technologies", "technologies", "trackers",
↳ "tracers", "personal data", "browsing data"
---
### IMPORTANT NOTE FOR VAGUE TERMS
- Contextual Evidence has STRICT priority over lexical vagueness.
- The model MUST evaluate the surrounding context BEFORE deciding whether a vague term is
↳ actually vague.
- The model MUST NOT rely on isolated keywords.
- Contextual grounding MUST be evaluated independently for EACH vague term.
- The presence of concrete examples for one vague term does NOT automatically resolve other
↳ vague terms appearing in the text.
A vague term is considered contextually grounded ONLY IF:
- After full contextual inspection, the nearby text contains CLEAR and SPECIFIC examples of
↳ collected data, identifiers, tracking signals, or device-related data.
The examples MUST be:
- Concrete and technical (e.g., IP address, cookie ID, precise geolocation, device identifier)

If the examples are:
- Vague, abstract, or high-level (e.g., "user data", "personal information", "various data",
↳ "online activity")
-> They DO NOT count as grounding evidence.
The examples section is illustrative and not exhaustive.
All grounding decisions MUST still satisfy the strict criteria above.
In such cases (when concrete and specific examples exist):
- the term MUST be treated as contextually grounded (NOT vague),
- even if the term itself is linguistically broad.
ONLY AFTER full contextual inspection,
AND only if NO concrete and specific grounding evidence exists,
MUST the term be classified as vague.
---
## 2. "vague_terms"
- List of all terms in "tracking_technique" which belong to the vague
  terms category and remain undefined after verifying contextual evidence.
---
## 3. "vague"
- `true` ONLY IF at least one term exists in "vague_terms"
- `false` otherwise
---
## 4. "cookies"
- `true` if the term "cookies" appears explicitly and affirmatively
- `false` otherwise
---
## 5. "Fingerprinting"
- `true` if at least one stateless technique is present in "tracking_technique"
- `false` otherwise
---
## 6. "explanation"
- Provide one short sentence explaining why the groups above were selected.
---
## OUTPUT FORMAT
Return a JSON object with this structure:
{
  "tracking_technique": ["string"],
  "vague_terms": ["string"],
  "vague": boolean,
  "cookies": boolean,

```

```

    "Fingerprinting": boolean,
    "explanation": "One short sentence."
  }

```

C Fingerprinting Techniques

Table 5. Number of unique websites where loaded third-party scripts accessed fingerprinting-related JavaScript properties after banner rejection, grouped by category.

Category	# websites
Navigator Identity Fingerprinting	9,099
Performance & Time Fingerprinting	8,674
Media I/O Fingerprinting	5,931
Screen/Display Fingerprinting	3,568
Keystroke Fingerprinting	325
DOM Layout Fingerprinting	111
Audio Fingerprinting	104
WebRTC Fingerprinting	21
Font Fingerprinting	13

Table 5 summarizes the prevalence of fingerprinting techniques based on JavaScript property access across websites. The most prevalent category is Navigator Identity Fingerprinting, observed on 9,099 sites. This class of techniques leverages the Navigator interface to access both explicit browser metadata and implicit capability disclosures exposed by the runtime environment, including properties such as `userAgent` and `hardwareConcurrency`. Media I/O fingerprinting also exhibited substantial adoption, appearing on 5,931 sites, indicating that APIs related to media devices and input/output channels are widely leveraged for tracking purposes. In comparison, Screen/Display and Performance & Time fingerprinting were observed on 3,568 and 8,674 sites, respec-

tively, reflecting the broad use of device and execution-context signals. More advanced fingerprinting techniques, such as WebRTC, remained comparatively fewer in number, appearing on 21 sites. This highlights that while basic fingerprinting vectors—particularly those based on navigator properties and screen characteristics—are widely deployed across the Web, there also exist non-trivial instances where more sophisticated fingerprinting techniques are employed, indicating that a subset of websites continues to adopt sophisticated fingerprinting techniques in addition to conventional tracking mechanisms.

D Impact of Website Ranking

We analyze the consistency of post-rejection fingerprinting behavior across disclosure categories and website popularity tiers to examine whether tracking practices vary among less popular websites. To study these variations, we analyze websites from the Tranco Top 100K and additionally include lower-ranked samples, consisting of 5K domains each from the ~250K, ~500K, ~750K, and ~1M ranking tiers.

As shown in Figure 7, websites belonging to the *Transparent Disclosure* category exhibit relatively consistent levels of post-rejection fingerprinting-related tracking across all ranking tiers. For websites categorized under *Partially Ambiguous Disclosure*, we consistently observe higher levels of fingerprinting activity compared to the Transparent Disclosure category across nearly all ranking tiers. Additionally, for the *Completely Ambiguous Disclosure* category, we observe an increasing trend in fingerprinting activity from the Top 100K to the ~250K tier, followed by a decrease across the ~750K and ~1M tiers. The tilde (~) denotes a lower band of 5K domains sampled from the bottom of each respective ranking threshold (i.e., ranks 245,001–250,000, 495,001–500,000, and so on). Note that at the 500K tier, we do not see Completely Ambiguous and No Disclosure group banners.

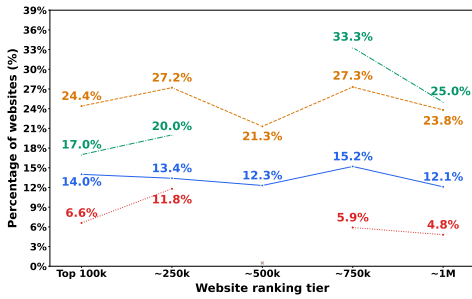
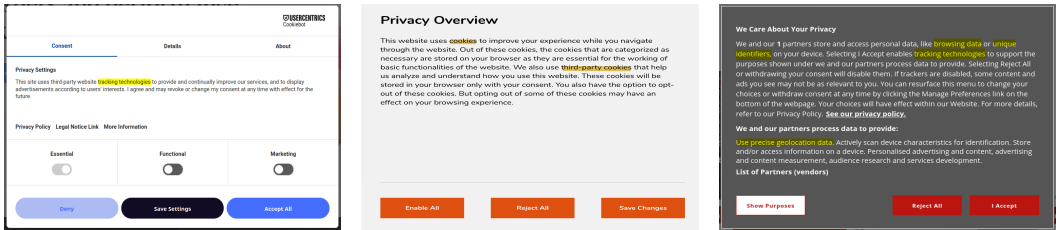


Fig. 7. Websites performing third-party fingerprinting despite setting no cookies following consent rejection, grouped by consent notice disclosure type across five Tranco ranking tiers: Top 100K, and the lower bands of the ~250K, ~500K, ~750K, and ~1M tiers. Transparent (●), Partially Ambiguous (●), Completely Ambiguous (●), No Disclosure (●).

These observations indicate that post-rejection fingerprinting behavior is not limited to highly popular websites but persists across multiple popularity tiers and disclosure categories. In particular, websites that employ partially or fully ambiguous disclosures consistently exhibit higher levels of fingerprinting-related activity than those that provide more transparent disclosures. This indicates that ambiguous disclosure practices are not confined to a specific segment of the web ecosystem and may continue to affect informed consent and transparency expectations even among lower-ranked and less frequently studied websites.

E Tracking Techniques Disclosed in Cookie Banners



(a) Banner with vague-only tracking techniques

(b) Banner with cookie-only tracking techniques

(c) Banner with fingerprinting and vague tracking techniques

Fig. 8. Examples of cookie banners with different tracking techniques

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